

Code:

```
#include <stdio.h>

#define MAX 100

int queue[MAX];
int front = -1;
int rear = -1;

void disp() {
    int i;
    if (front == -1) return;
    for(i=front;i<=rear;i++) {
        printf("%d ", queue[i]);
    }
}

void enqueue(int data) {
    if (rear == MAX - 1) return;
    if (front == -1) front = 0;
    queue[++rear] = data;
}

int main() {
    int size;
    if (scanf("%d", &size) != 1) return 0;

    for (int j = 0; j < size; j++) {
        int data;
        scanf("%d", &data);

        disp();
        printf("Enqueuing %d\n", data);
        enqueue(data);
    }

    // Final state of queue
    for (int i = front; i <= rear; i++) {
        printf("%d", queue[i]);
        if (i < rear) printf(" ");
    }
}
```

```
    printf("\n");  
  
    return 0;  
}
```

Input:

```
5  
5 6 3 1 2
```

Output:

```
Enqueuing 5  
5 Enqueuing 6  
5 6 Enqueuing 3  
5 6 3 Enqueuing 1  
5 6 3 1 Enqueuing 2  
5 6 3 1 2
```